

SUZY TANG

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SUMMARY

- Quantitative Analyst with 9 years of experience across FX execution analytics, statistical modelling, and production machine learning.
- At HSBC, built TCA frameworks and execution-quality models on large-scale FX transaction data, surfacing recoverable execution-slippage opportunities and materially reducing abnormal-spread incidents across G10 APAC currency pairs.
- Independent systematic-FX research: G10 carry strategy with volatility-aware risk overlay; HMM-based volatility-regime detection on high-frequency EUR/USD; cointegration-based statistical arbitrage on G10 FX pairs.
- Completing MSc Data Science (UTS, expected Dec 2026); strong in Python, SQL, time-series modelling (ARIMA/GARCH), statistical ML, and Spark-scale data pipelines.

EXPERIENCE

Applied Data Science Research Assistant | *University of Technology Sydney* Sep 2024 – Present

- Lead data science workstream of an applied research project in collaboration with a government agency and real-estate industry partner.
- Built end-to-end Python pipelines for statistical modelling and forecasting of property-market data; delivered decision-support outputs to academic and industry stakeholders.

Senior Analyst, FX Analytics | *HSBC* | *Hong Kong* Jun 2021 – Jul 2024

- Owned development of transaction cost analysis and execution-quality analytics on large-scale FX transaction data, supporting trading, product, risk, and engineering stakeholders across APAC markets.
- Developed GARCH-based and time-series models to characterise spread dynamics and detect anomalies in execution behaviour across volatility regimes; models materially reduced abnormal-spread incidents.
- Modelled interactions between spot dynamics, spread widening, and liquidity conditions to identify drivers of execution slippage; outputs directly informed pricing calibration and execution-timing decisions.
- Analysed hundreds of millions of FX transactions to characterise flow behaviour across market regimes, improving execution-quality benchmarking and spread calibration across G10 APAC pairs.
- TCA framework surfaced recoverable execution-slippage opportunities; analysis used by trading desks to optimise execution strategy.

Data Scientist | *Huawei* | *China* Jan 2018 – May 2021

- Built and deployed production ML models serving multilingual users at scale; owned model performance, low-latency inference, and degradation monitoring.
- Designed large-scale A/B experimentation and statistical evaluation frameworks to measure treatment effects and optimise model-driven decisions across multi-region datasets.
- Built user-behaviour models on Spark/Hadoop pipelines processing hundreds of millions of records.

Data Scientist | *Nestlé* | *China* Sep 2016 – Jan 2018

- Automated ETL pipelines and reporting (Python/SQL), reducing manual reporting effort by approximately 20%.
- Built regional KPI dashboards enabling faster identification of underperforming markets.

SELECTED QUANTITATIVE PROJECTS

- **Systematic G10 FX Carry Strategy:** Built a carry-strategy framework in Python with a volatility-targeted risk overlay and transaction-cost sensitivity analysis; evaluated performance persistence across market regimes (2010–2024 backtest).
- **FX Volatility Regime Detection:** Built a Hidden Markov Model on high-frequency EUR/USD data to classify volatility regimes; identified structurally distinct market states and analysed regime-dependent spread behaviour and momentum-signal stability.
- **Statistical Arbitrage:** Built a cointegration-based pairs trading strategy on G10 FX pairs; implemented ADF/Johansen testing, rolling hedge-ratio estimation, and mean-reversion entry/exit signals.
- **Portfolio Construction:** Built mean-variance and risk-parity portfolio allocation frameworks; evaluated turnover, concentration risk, and risk-adjusted performance under different volatility regimes.

EDUCATION

- MSc Data Science — University of Technology Sydney | Expected Dec 2026
- Coursework: Statistical Machine Learning, Time Series Analysis, Advanced Data Analytics, Big Data Engineering.

SKILLS

Languages: Python (Pandas, NumPy, SciPy, scikit-learn), SQL, R, C++

Quant / ML: Time-series modelling (ARIMA, GARCH), regime-switching / HMM, statistical arbitrage, XGBoost, logistic regression, A/B testing

Domain: FX execution analytics, transaction cost analysis (TCA), market microstructure, volatility modelling

Tools: Spark, Hadoop, Kubernetes, TensorFlow, HSBC Evolve